

SDG 15 Life on Land Evidence Compendium 2025

Mariano Marcos State University · THE Sustainability Impact Ratings 2027

Compiled from MMSU, DOST-PCAARRD and news sources for the 2027 data submission (due 9 November 2026). Each section corresponds to one THE SDG 15 indicator: 2025 evidence first, then policies and programmes that continue from earlier years and into 2026.

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MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.1 · Events about sustainable use of land

Support or organise events aimed to promote conservation and sustainable utilisation of the land, including forests and wild land.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

Evidence from 2025

CAFSD Graduands Plant Native Trees and Bamboo in the University Forest Reserves



CAFSD Graduating Class 2025 "Bamboo and Tree Planting Activity," University Forest Reserves, City of Batac, 2 July 2025. (Photo: MMSU STRATCOM)



Bamboo planting site in the University Forest Reserve, Brgy. Mabaleng, 2 July 2025. (Photo: MMSU STRATCOM)

Some 230 graduating students of the College of Agriculture, Food and Sustainable Development (CAFSD) of Mariano Marcos State University (MMSU) planted trees and bamboo in the University Forest Reserves in Barangays Payao and Mabaleng, City of Batac, on July 2, 2025. The activity was led by CAFSD OIC-Dean Charlie Batin and the 2025 CAFSD Graduating Class Student Council, with Dr. Joselito Rosario, Director for Agribusiness, and Mr. Bonne Jovie Orteza of the MMSU Bamboo Research and Development Center. MMSU President Virgilio Julius P. Manzano Jr. commended the class for leaving a lasting green footprint that will further preserve the forest areas managed by the university, and Mr. Hashler Garalde, a CAFSD alumnus representing the Provincial Government of Ilocos Norte, supported the college's sustainability efforts.

The graduands planted 120 seedlings of guijo and yakal, native dipterocarp timber species raised at the MMSU Central Nursery, in the Payao Forest Reserve, and 130 propagules of kawayan tinik, machiku and giant bamboo from the MMSU Bamboo Nursery in Mabaleng.

Source

- MMSU News, "CAFSD graduands plant trees, bamboos in Batac forest reserves," 4 July 2025 https://www.mmsu.edu.ph/article/news/2468/cafsd-graduands-plant-trees-bamboos-in-batac-forest-reserves
- MMSU News, "BS Forestry seniors leave green footprints with pre-graduation tree planting," 11 July 2024 https://www.mmsu.edu.ph/article/news/2065/bs-forestry-seniors-leave-green-footprints-with-pre-graduation-tree-planting

MMSU Plants 250 Mangrove Propagules in Currimao on World Environment Day



Mangrove planting for World Environment Day, Brgy. Maglaoi, Currimao, June 2025. (Photo: MMSU STRATCOM)



MMSU, LGU Currimao, DENR and fisherfolk volunteers at the planting site, June 2025. (Photo: MMSU STRATCOM)

In June 2025, in celebration of World Environment Day, Mariano Marcos State University planted 250 mangrove propagules along the coastal area of Barangay Maglaoi, Currimao, Ilocos Norte, to restore degraded shorelines, enhance climate resilience and promote sustainable coastal resource management. The activity was spearheaded by the MMSU Sustainability and Futures Thinking (SFT) Office with the Local Government Unit of Currimao and led by Dr. Mee Jay Domingo, Vice President for Planning and Strategic Foresight.

MMSU faculty, education support personnel and students joined representatives of the Department of Environment and Natural Resources, barangay officials, members of local fisherfolk organizations and community members. SFT Director Dr. Arlene Gonzales explained that mangroves were chosen because they act as natural filters that intercept plastic waste from land before it reaches the sea, while absorbing carbon, protecting shorelines from erosion and providing habitat for marine life.

Source

- MMSU News, "MMSU plants 250 mangrove propagules in Currimao," 11 June 2025 https://www.mmsu.edu.ph/news/mmsu-plants-250-mangrove-propagules-in-currimao

Green Wall Program: Coconut Seedlings for the Currimao Coastline



Coconut seedling planting along the Currimao beachfront under the MMSU Green Wall Program, 24 July 2025. (Photo: MMSU)

On July 24, 2025, MMSU planted coconut seedlings along the Currimao beachfront under its Green Wall Program, with 1,000 seedlings donated by the Philippine Coconut Authority (PCA), to promote environmental sustainability and coastal resilience amid growing climate threats. The activity was held in celebration of the birthday of CHED Chairperson Shirley C. Agrupis, Chair of the MMSU Board of Regents.

MMSU President Virgilio Julius P. Manzano Jr., PCA Ilocos Norte Agriculturist and In-Charge Rose Ann Cabaloan, and university officials and employees took part. CASAT OIC-Dean Pilar Carolyn Pascual highlighted the university's commitment to protecting coastal ecosystems through the Green Wall project.

Source

- MMSU News, "MMSU celebrates CHED Chair Agrupis' birthday with coconut planting activity," 28 July 2025 https://www.mmsu.edu.ph/news/mmsu-celebrates-ched-chair-agrupis-birthday-with-coconut-planting-activity

MMSU Promotes Regenerative Farming at the 2025 NSTW Forum



Forum series of the 2025 National Science and Technology Week, Northwestern University, Laoag City, 19 November 2025. (Photo: MMSU)

On November 19, 2025, MMSU Research Director Dr. Dionisio Bucao served as a resource speaker at the forum series of the 2025 National Science and Technology Week held at Northwestern University, Laoag City. He presented regenerative farming approaches that restore soil health, biodiversity and ecosystem function, and emphasized carbon sequestration, improved soil fertility, water conservation and climate resilience through natural processes, as well as how regenerative farming differs from conventional practice.

Source

- MMSU News, “MMSU researchers share expertise on smart agriculture in NSTW forum,” 20 November 2025 https://www.mmsu.edu.ph/news/mmsu-researchers-share-expertise-on-smart-agriculture-in-nstw-forum

President’s Farm Field Day Showcases Climate-Resilient Rice Varieties



President’s Farm technology demonstration of inbred and hybrid rice varieties, 28 October 2025. (Photo: MMSU)



Farmers, LGU technicians and students at the President’s Farm Field Day, in front of CAFSD. (Photo: MMSU)

On October 28, 2025, the MMSU Extension Directorate, with the Philippine Rice Research Institute (PhilRice) Batac Branch, held the President’s Farm Field Day: A Technology Showcase in front of the College of Agriculture, Food and Sustainable Development. Farmers and farmer-leaders, agricultural technicians from Ilocos Norte local government units, BS Agriculture students, and MMSU and PhilRice rice specialists viewed eight rice varieties grown under wet-season conditions.

The varieties included four newly released by PhilRice, two public hybrids, and two suited to saline and rainfed conditions, among them NSIC Rc 686 (Salinas 39) and NSIC Rc 650 (Sahod Ulan 36), giving farmers options for saline-affected and rainfed fields. Speakers included MMSU President Dr. Virgilio Julius P. Manzano Jr., Extension Director Dr. Marlowe Aquino, and former MMSU President and PhilRice Executive Director Dr. Santiago R. Obien.

Source

- MMSU News, “MMSU President’s Farm showcases 8 rice varieties,” 4 November 2025 https://www.mmsu.edu.ph/article/news/2665/mmsu-presidents-farm-showcases-8-rice-varieties

2025 International Day for Biological Diversity Wildlife Photo Contest



Contest poster with the animal-photography ethical guidelines (deadline 30 May 2025).



Winner Chris Jeremiah C. Antonio receives his certificate from Dr. Arlene L. Gonzales, Chief of Sustainability and Futures Thinking.

MMSU, through its Office of Sustainability and Futures Thinking, marked the 2025 International Day for Biological Diversity with a Wildlife Photo Contest that invited the university community to document local wildlife and promote awareness of biodiversity conservation. Entrants followed animal-photography ethical guidelines: avoiding flash and loud noises, never using bait or food lures, following local laws and warning signs, keeping clear of nesting sites and sensitive habitats, and captioning photographs honestly. Entries were received until May 30, 2025.

Chris Jeremiah C. Antonio, a third-year BA Communication student, was named winner and received a ₱1,000 gift voucher and a certificate from Dr. Arlene L. Gonzales, Chief of Sustainability and Futures Thinking.

Source

- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

Graduation Tree Planting Is a Standing MMSU Programme, 2024 to 2026

Continuing implementation · July 2024, July 2025, June 2026



CAFSD Graduating Class 2026 enrichment planting at the MMSU Forest Reserve, Payao, 1 June 2026. (Photo: MMSU STRATCOM)

Graduating classes of the College of Agriculture, Food and Sustainable Development (CAFSD) of Mariano Marcos State University (MMSU) plant trees before they graduate, and the planting has been repeated every year. On July 9, 2024, the BS Forestry graduating class, led by CAFSD OIC-Dean Charlie Batin, enriched the Salinged Park, the university's ecological park, with igem-dagat, balitbitan, bitaog, Cassia javanica and banaba. On July 2, 2025, 230 CAFSD graduands planted guijo, yakal and bamboo in the University Forest Reserves in Payao and Mabaleng, City of Batac.

On June 1, 2026, the CAFSD Graduating Class of 2026, led by its Student Council with the chairs and faculty of the Departments of Environmental Science and Forestry, carried out an enrichment planting at the MMSU Forest Reserve in Payao, which the CAFSD Batch 2025 had previously planted with native seedlings. They planted 50 banaba, 30 yakal saplungan and 20 guijo seedlings at the onset of the rainy season, hardened in the MMSU Forest Tree Nursery, species chosen because they improve soil quality, conserve biodiversity and absorb and store carbon dioxide. Dr. Arlene Gonzales, Director of the Sustainability and Futures Thinking Office, represented the University President and commended the college for sustaining the initiative. The activity follows Republic Act 10800, the Graduation Legacy for the Environment Act, which requires students to plant trees before graduation.

Source

- MMSU News, "BS Forestry seniors leave green footprints with pre-graduation tree planting," 11 July 2024 https://www.mmsu.edu.ph/article/news/2065/bs-forestry-seniors-leave-green-footprints-with-pre-graduation-tree-planting
- MMSU News, "CAFSD graduands plant trees, bamboos in Batac forest reserves," 4 July 2025 https://www.mmsu.edu.ph/article/news/2468/cafsd-graduands-plant-trees-bamboos-in-batac-forest-reserves
- MMSU News, "CAFSD graduands grow native trees at MMSU forest reserve," 5 June 2026 https://www.mmsu.edu.ph/news/cafsd-graduands-grow-native-trees-at-mmsu-forest-reserve

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.3 · Maintain and extend current ecosystems' biodiversity

Work directly to maintain and extend existing ecosystems and their biodiversity, of both plants and animals, especially ecosystems under threat.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

Evidence from 2025

Enriching MMSU's 1,068-Hectare Forest Reserves with Native Species



Bamboo planting site in the University Forest Reserve, Brgy. Mabaleng, 2 July 2025. (Photo: MMSU STRATCOM)

MMSU manages Forest Reserve Areas in Barangays Lubbot and Payao, City of Batac, covering a total of 1,068.13 hectares, where the university's forestry and agroforestry research is set up. The university maintains and extends these ecosystems with planting stock raised in its own nurseries, the MMSU Central Nursery and the MMSU Bamboo Nursery.

On July 2, 2025, 230 CAFSD graduating students enriched the reserves with 120 seedlings of the native dipterocarps guijo and yakal in Payao and 130 propagules of kawayan tinik, machiku and giant bamboo in Mabaleng. The activity was led by CAFSD OIC-Dean Charlie Batin with the MMSU Bamboo Research and Development Center.

Source

- MMSU Research Directorate, Station Profile
  https://research.mmsu.edu.ph/about/station-profile
- MMSU News, "CAFSD graduands plant trees, bamboos in Batac forest reserves," 4 July 2025
  https://www.mmsu.edu.ph/article/news/2468/cafsd-graduands-plant-trees-bamboos-in-batac-forest-reserves

Conserving Ilocos Indigenous Food Plants In Situ, Ex Situ and Circa Situ



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

Through its Indigenous Food Plants (IFP) Program, MMSU conserves the wild and neglected food plants of Ilocos Norte, 46 of which were documented by the program in seven towns of the province. The program follows a triad of conservation approaches: an IFP Botanical Garden for ex situ conservation, a himbabao (alukon) genebank at the MMSU Forest Reserve, circa situ cultivation of the wild yams kamangeg and dolian, and the integration of IFPs into community foodscapes.

In October 2025, in celebration of World Food Day, the MMSU Extension and Research Directorates established the second batch of Community Foodscapes and Seedbanks for Indigenous Food Plants with local government units and schools, at Eladio V. Barangan Memorial Elementary School in San Nicolas, Ilocos Norte. The foodscape sites serve as seedbanks and learning gardens, and partner communities receive planting materials and technical assistance to cultivate kapaskapas, alokon, pako, dalayap, wild singkamas and roselle. The new partners join existing foodscape sites in Sarrat and Batac.

The IFP Program was adopted by the Regional Development Council of Region 1 as its first research and development initiative, and a study on the program by Dr. Menisa A. Antonio and co-authors won Best Paper in the Sustainable and Smart Agri-Aquaculture and Climate Resilience category of the SUC-ACAP 2025 Biennial Conference, held August 27–29, 2025, in Tagbilaran City, Bohol.

Source

- MMSU Research Directorate, "MMSU research on indigenous food plants wins best paper at SUC-ACAP 2025 conference" https://research.mmsu.edu.ph/news/details/mmsu-research-on-indigenous-food-plants-wins-best-paper-at-suc-acap-2025-conference
- MMSU Extension Directorate, "MMSU expands indigenous food plant seedbanks and foodscapes in LGUs," October 2025 https://extension.mmsu.edu.ph/news/show/389/mmsu-expands-indigenous-food-plant-seedbanks-and-foodscapes-in-lgus
- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

Nature-based Solutions for Mangrove Ecosystems in Badoc and Currimao



Mangrove site, from the gallery of the October 2025 seminar-workshop of the NbS mangrove project. (Photo: MMSU)



Dr. Arlene L. Gonzales at the MMSU–DENR Region 1 dialogue on mangrove conservation, ILAARRDEC.

Throughout 2025, MMSU implemented the DOST-PCAARRD-funded project "Nature-based Solutions (NbS) for Sustainable Mangrove Management and Resilient Coastal Communities Through Innovative Green Engineering Approach," led by Dr. Arlene L. Gonzales, in Badoc and Currimao, Ilocos Norte, with DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the LGUs of Badoc and Currimao. The project uses green engineering and locally available, ecologically friendly materials, including bamboo-engineered structures that protect young mangroves and improve their survival.

In 2025, MMSU and DENR Region 1 met at ILAARRDEC to advance the work, with DENR pledging technical support and active participation in restoration, monitoring and evaluation, and the project trained representatives of Currimao, Pasuquin, Badoc and Burgos in mangrove propagation, management and conservation on October 27–28, 2025.

Source

- MMSU News, "MMSU trains Ilocos Norte coastal towns on mangrove propagation and management," 5 November 2025 https://www.mmsu.edu.ph/article/news/2670/mmsu-trains-ilocos-norte-coastal-towns-on-mangrove-propagation-and-management
- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand



MMSU, Rotary Club of Laoag Sunshine and LGU-Badoc plant mangrove propagules in Brgy. Pagsanahan, Badoc, 6 December 2025. (Photo: MMSU)



Participants of the mangrove enhancement and growing activity in Badoc, 6 December 2025. (Photo: MMSU)

On December 6, 2025, Mariano Marcos State University, the Rotary Club of Laoag Sunshine and the Local Government Unit of Badoc planted 280 mangrove propagules in Barangay Pagsanahan, Badoc, Ilocos Norte, to restore a mangrove stand extensively damaged by past typhoons and to ensure the continuity of ecological succession at the site.

The activity was led by MMSU President Virgilio Julius P. Manzano Jr., with the LGU of Badoc headed by Hon. Virgilio M. Calajate and the Rotary Club of Laoag Sunshine under Atty. Zsa Zsa Raval. MMSU's Vice President for Resource Generation and its Directors for Extension, Curriculum Development and Innovative Learning, and Student Affairs and Development also took part.

Source

- MMSU News, "MMSU ties up with Rotary Club of Laoag Sunshine for mangrove planting in Badoc town," 11 December 2025 https://www.mmsu.edu.ph/news/mmsu-ties-up-with-rotary-club-of-laoag-sunshine-for-mangrove-planting-in-badoc-town

A New Bambusetum in Barangay Aglipay, Pinili



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

During a two-day bamboo production training in 2025, the MMSU Extension Directorate and CAFSD, with Barangay Aglipay, Pinili, and the support of ILAARRDEC, worked with 46 farmers, local leaders and community members to establish a new bambusetum, a collection of bamboo species, in Barangay Aglipay. The training covered bamboo propagation, nursery establishment and plantation management.

Source

- MMSU News, “Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili,” 3 September 2025 https://www.mmsu.edu.ph/news/sustainable-livelihood-mmsu-ilaarrdec-hold-bamboo-training-in-pinili

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

Mangrove Nature-based Solutions Project Runs from 2024 to 2026

Continuing implementation · 1 May 2024 to 30 April 2026



Project Leader Dr. Arlene L. Gonzales presents the project at its launch, MMSU Review Center, 27 June 2024. (Image credit: TTPD, DOST-PCAARRD)



The MMSU project team with DOST-PCAARRD, DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the LGUs of Badoc and Currimao at the launch. (Image credit: TTPD, DOST-PCAARRD)

The DOST-PCAARRD-funded project "Nature-based Solutions (NbS) for Sustainable Mangrove Management and Resilient Coastal Communities Through Innovative Green Engineering Approach," led by Dr. Arlene L. Gonzales of MMSU, was launched at the MMSU Review Center on June 27, 2024, and runs from May 1, 2024 to April 30, 2026 in Badoc and Currimao, Ilocos Norte. DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the local government units of Badoc and Currimao support the project.

The 2025 activities of the project, including the dialogue with DENR Region 1 and the October 2025 mangrove propagation seminar-workshop, are described under Indicators 15.2.3 and 15.2.4.

Source

- DOST-PCAARRD, "DOST-PCAARRD, MMSU launch nature-based solutions project on mangrove management in Ilocos Norte," 7 August 2024
  https://www.pcaarrd.dost.gov.ph/index.php/quick-information-dispatch-qid-articles/dost-pcaarrd-mmsu-launch-nature-based-solutions-project-on-mangrove-management-in-ilocos-norte

Bambusetums Create Living Collections for Bamboo Conservation and Learning

Continuing implementation · September 2025 and September 2026



Launch of the bambusetum at the MMSU Dingras Campus, 30 September 2026. (Photo: MMSU STRATCOM)

In celebration of Bamboo Month, the MMSU Dingras Campus, through CAFSD, launched its bambusetum on September 30, 2026, to advance bamboo research, education and sustainable use. Students, faculty, field workers and stakeholders joined the planting, where For. Charlie Maine Cacactin demonstrated the proper planting of various bamboo species, and members of the Association of Young Agriculturists, Green Triangle Organization, Health and Wellness Club and Junior Foresters' Association supported the activity. The bambusetum is the second in the university, after the one at the Main Campus.

The bambusetum will serve as a living collection of bamboo species and a learning site for teaching, research, technology development and community extension, showcasing bamboo's ecological and socioeconomic benefits for food security, livelihood, environmental conservation and climate resilience. It follows the bambusetum that training participants established in Barangay Aglipay, Pinili, in 2025. CAFSD Dean Charlie Batin said the work responds to environmental problems such as climate change.

Source

- MMSU News, "MMSU CAFSD-Dingras launches bambusetum to advance bamboo research and education," 2 October 2026 https://www.mmsu.edu.ph/article/news/3206/mmsu-cafsd-dingras-launches-bambusetum-to-advance-bamboo-research-and-education
- MMSU News, "Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili," 3 September 2025 https://www.mmsu.edu.ph/news/sustainable-livelihood-mmsu-ilaarrdec-hold-bamboo-training-in-pinili

Biodiversity Conservation Learning Center and Leisure Park Is Enriched with 28 Native Tree Species

Continuing implementation · since 2023

On January 13, 2023, MMSU planted 120 more seedlings of 28 native tree species and two bamboo species at the Biodiversity Conservation Learning Center and Leisure Park on the Main Campus, led by then President Shirley C. Agrupis with administrators, faculty, staff and students. The park is used as a field laboratory for classes and was identified for development into a health and wellness tourism hub, a plan endorsed by the provincial government and aligned with the Health and Wellness Complex of the College of Medicine.

The planting continues a tradition that began in 2017 with the Green Wall project on the Batac campus and extends to the Nagbacsan site and the Currimao, Dingras and Laoag campuses, as the university's contribution to the provincial "Green Wall of Ilocos Norte" and the National Greening Program of the Department of Environment and Natural Resources.

Source

- MMSU News, "More trees planted at MMSU to establish 'health tourism' hub," 13 January 2023 https://www.mmsu.edu.ph/news/more-trees-planted-at-mmsu-to-establish-health-tourism-hub

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.4 · Educational programmes on ecosystems

Offer educational programmes on ecosystems (looking at wild flora and fauna) for local or national communities.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

MMSU Trains Coastal Towns on Mangrove Propagation, Management and Conservation



Representatives of Currimao, Pasuquin, Badoc and Burgos at the Seminar-Workshop on Mangrove Propagation, 27–28 October 2025. (Photo: MMSU)



Hands-on mangrove nursery and propagation practice during the seminar-workshop. (Photo: MMSU)

MMSU conducted a two-day Seminar-Workshop on Mangrove Propagation on October 27–28, 2025 at its Center for Flexible Learning for representatives of the coastal municipalities of Currimao, Pasuquin, Badoc and Burgos, Ilocos Norte, with officials of DENR-PENRO in attendance. The training was organized under the DOST-PCAARRD-funded nature-based solutions mangrove project led by Dr. Arlene Gonzales.

Sessions covered the importance of mangrove ecosystems, nature-based solutions for conservation, legal frameworks for mangrove protection, and the roles of the academe and local government units in community-based mangrove management. Participants were also trained in

nursery management and advanced propagation techniques for Avicennia and Sonneratia species and other local mangrove varieties. The Pedada People's Organization shared its experience in establishing community-based mangrove conservation.

Source

- MMSU News, "MMSU trains Ilocos Norte coastal towns on mangrove propagation and management," 5 November 2025 https://www.mmsu.edu.ph/article/news/2670/mmsu-trains-ilocos-norte-coastal-towns-on-mangrove-propagation-and-management

Farmers in Sarrat Trained to Propagate the Indigenous Alukon



Farmers and local officials trained on alukon (himbabao) propagation, Brgy. 11 Sta. Rosa, Sarrat, 2 May 2025. (Photo: MMSU)

On May 2, 2025, the MMSU Research and Extension Directorates trained 64 farmers in Barangay 11 Sta. Rosa, Sarrat, Ilocos Norte, on the propagation of alukon (himbabao), a Philippine indigenous vegetable tree. Barangay officials, members of the Task Force Dagger Special Operations Foundation and representatives of the local government unit also attended. The training was held in response to requests from the community.

Resource speakers Dr. Menisa Antonio, Engr. Josefa Pugat and Mr. Jayson Rasay of the Research Directorate presented recommended propagation strategies to establish sustained alukon production in the community and broaden the use of the crop for food and other applications.

Source

- MMSU News, "MMSU trains Sarrat farmers on alukon propagation," 9 May 2025 https://www.mmsu.edu.ph/news/mmsu-trains-sarrat-farmers-on-alukon-propagation

MMSU Brings Bamboo and Alukon Propagation to the 2025 Siwawer Farmers' Festival



MMSU Information and Technology Caravan at the 2025 Siwawer Farmers' Festival, Vintar, 3–4 December 2025. (Photo: MMSU STRATCOM)

Through its Information and Technology Caravan, MMSU's Research and Extension Directorates joined the 2025 Siwawer Farmers' Festival and Agro-Trade Fair at the Vintar Public Auditorium on December 3–4, 2025, with an exhibit of university technologies and a series of expert lectures for farmers.

Forester Shena Rose Garcia introduced sustainable bamboo production using MMSU's one-node cutting technology, and the second day featured a demonstration on the propagation of alukon, an indigenous vegetable tree, facilitated by Ms. Josefa Pugat. Other lectures covered high-yielding glutinous corn varieties, garlic disease management and black garlic.

Source

- MMSU News, "MMSU IT caravan goes to Vintar," 15 December 2025 https://www.mmsu.edu.ph/news/mmsu-it-caravan-goes-to-vintar

Community Foodscapes as Learning Gardens for Indigenous Plants



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

In October 2025, in celebration of World Food Day, the MMSU Extension and Research Directorates established the second batch of Community Foodscapes and Seedbanks for Indigenous Food Plants at Eladio V. Barangan Memorial Elementary School, San Nicolas, Ilocos Norte. Barangay officials of San Roque and Parut, Vintar, agricultural extension workers, school leaders and community representatives took part.

The foodscape sites serve as seedbanks and learning gardens that encourage knowledge sharing and sustainable farming practices. Communities receive planting materials and technical assistance to cultivate indigenous crops such as kapaskapas, alokon, pako, dalayap, wild singkamas and roselle, conserving indigenous food resources across Ilocos Norte.

Source

- MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025 https://extension.mmsu.edu.ph/news/show/389/mmsu-expands-indigenous-food-plant-seedbanks-and-foodscapes-in-lgus
- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.5 · Sustainable management of land for agriculture (educational outreach)

Offer educational programme/outreach for local or national communities on sustainable management of land for agriculture.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

School-on-the-Air Brings Regenerative Vegetable Farming to 1,000 Ilocanos



MMSU School-on-the-Air (Eskuela iti Tangatang) season on regenerative vegetable farming, June–December 2025. (Image: MMSU)

A new season of the MMSU School-on-the-Air (SOA), the university's radio-based educational program, opened on June 24 and 25, 2025 with 1,000 enrollees from the 22 municipalities and cities of Ilocos Norte. The season focused on regenerative vegetable farming for enhanced and sustainable production, with lessons on climate-smart practices and off-season vegetable seeding.

Vegetable growers, rural improvement club members, agricultural extension workers and students followed the lessons, broadcast from June to December 2025 over DWFB Radyo Pilipinas Laoag, DWCI Radyo Adjo Piddig and DWNI Radyo Karruba Burgos. MMSU ran the program with the

Provincial Government of Ilocos Norte, the Office of the Provincial Agriculturist, local government units, the Ilocos Agriculture, Aquatic and Natural Resources Research and Development Consortium, and the High-Value Crops Division of Ilocos Norte.

Source

- MMSU News, "MMSU draws 1K enrollees for SOA program on sustainable vegetable farming," 26 June 2025 https://www.mmsu.edu.ph/news/mmsu-draws-1k-enrollees-for-soa-program-on-sustainable-vegetable-farming

Parambak ditoy MMSU 2025 Showcases Extension Technologies for Farmers



7th Parambak ditoy MMSU, in front of CAFSD, 14 February 2025. (Photo: MMSU STRATCOM)

On February 14, 2025, to close its 47th Foundation Anniversary, MMSU held the 7th Parambak ditoy MMSU in front of the College of Agriculture, Food and Sustainable Development, highlighting the impact of its extension and research programmes on community development and agricultural innovation. Thirty-seven exhibitors took part, including learning sites of the Agricultural Training Institute–Ilocos Region, Magsasaka Siyentista (farmer scientists), Balik-Probinsya beneficiaries and local government units, and the Research Directorate gave a speed talk on MMSU technologies such as off-season tomato and shallot production, Bolinao chicken, indigenous vegetables and black garlic.

Participants harvested fresh vegetables at the President's Farm, 17 new partners of the Extension Directorate received certificates of appreciation, and beneficiaries shared how MMSU's technologies had supported sustainable agriculture. Attendees included the Provincial Government of Ilocos Norte, LGUs, non-government organisations, farmer leaders, entrepreneurs and MMSU-assisted beneficiaries. Incoming President Virgilio Julius P. Manzano Jr. urged the units concerned to sustain sustainable farming practices.

Source

- MMSU Extension Directorate, "'Parambak' caps MMSU@47 celebration," February 2025 https://extension.mmsu.edu.ph/news/show/368/parambak-caps-mmsu-at-47-celebration

Bamboo Propagation and Plantation Management Training in Pinili



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

The MMSU Extension Directorate and the College of Agriculture, Food and Sustainable Development, in partnership with Barangay Aglipay, Pinili, Ilocos Norte, and with the support of the Ilocos Agriculture, Aquatic and Natural Resources Research and Development Consortium (ILAARRDEC), conducted a two-day training on bamboo production in 2025.

The training equipped 46 farmers, local leaders and community members with practical skills in bamboo propagation, nursery establishment and plantation management. Lectures covered the prospects of the bamboo industry and MMSU's propagation methods, and participants helped establish a new bambusetum, a collection of bamboo species, in Barangay Aglipay. Lectures and demonstrations were given by bamboo experts Mr. Charlie Maine Cacactin of CAFSD and Mr. Bonne Jovie Orteza of MMSU. Pinili Municipal Agriculturist Philip Galang, representing Mayor Rommel T. Labasan, thanked MMSU and ILAARRDEC, and the project was championed in the community by Pinili's Magsasaka Siyentista, Mr. Reynaldo Antalan.

Source

- MMSU News, "Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili," 3 September 2025 https://www.mmsu.edu.ph/news/sustainable-livelihood-mmsu-ilaarrdec-hold-bamboo-training-in-pinili

Vermicast Facility Helps Pasuquin Onion Farmers Restore Degraded Soils



Pilot vermicomposting facility of Pasuquin white onion farmers, 2025. (Photo: MMSU STRATCOM)

In 2025, white onion farmer-leaders Mr. Marcos Palacay and Mr. Markflor Valdez of the Ayat ti Pasuquin Farmer and Irrigators Association established a pilot vermicomposting facility in Pasuquin, Ilocos Norte, under the Agri-Tech Facilitator Program of the Farmers Entrepreneurship Program of MMSU and the Jollibee Group Foundation. The facility uses African night crawlers (Eudrilus eugeniae) to convert biodegradable farm waste into vermicast, an organic fertilizer that improves soil structure, stimulates microbial activity and reduces the need for chemical inputs.

The project responds to soil degradation from prolonged reliance on chemical fertilizers, which the farmers linked to poor soil conditions and low seedling survival, and promotes sustainable solutions for white onion farming in the area.

Source

- MMSU News, “Vermicast facility established in Pasuquin as MMSU, Jollibee Group aid onion farmers,” 1 August 2025 https://www.mmsu.edu.ph/news/vermicast-facility-established-in-pasuquin-as-mmsu-jollibee-group-aid-onion-farmers

Pasuquin Onion Farmers Trained on Good Agricultural Practices



Pre-Dispersal Training on White Onion Production, Pasuquin Municipal Hall, 12 November 2025. (Photo: MMSU STRATCOM)

On November 12, 2025, the Municipality of Pasuquin, MMSU and the Jollibee Group Foundation conducted a Pre-Dispersal Training on White Onion Production at the Pasuquin Municipal Hall for farmer beneficiaries of the San Juan Icas Manuel Farmers' Association. The training covered white onion production technology, good agricultural practices (GAP) and proper farm recording for sustainable and profitable farming.

Farmer participation in the project has doubled from 15 to 30 beneficiaries, with a collective production target of 200 tons of white onion for 2025. The Office of the Provincial Agriculturist provides seeds and planting materials, complementing MMSU's technical support and capacity development.

Source

- MMSU News, "Pasuquin onion farmers trained by MMSU and Jollibee Group Foundation," 18 November 2025 https://www.mmsu.edu.ph/news/pasuquin-onion-farmers-trained-by-mmsu-and-jollibee-group-foundation

President's Farm Field Day Showcases Climate-Resilient Rice Varieties



President's Farm technology demonstration of inbred and hybrid rice varieties, 28 October 2025. (Photo: MMSU)



Farmers, LGU technicians and students at the President's Farm Field Day, in front of CAFSD. (Photo: MMSU)

On October 28, 2025, the MMSU Extension Directorate, with the Philippine Rice Research Institute (PhilRice) Batac Branch, held the President's Farm Field Day: A Technology Showcase in front of the College of Agriculture, Food and Sustainable Development. Farmers and farmer-leaders, agricultural technicians from Ilocos Norte local government units, BS Agriculture students, and MMSU and PhilRice rice specialists viewed eight rice varieties grown under wet-season conditions.

The varieties included four newly released by PhilRice, two public hybrids, and two suited to saline and rainfed conditions, among them NSIC Rc 686 (Salinas 39) and NSIC Rc 650 (Sahod Ulan 36), giving farmers options for saline-affected and rainfed fields. Speakers included MMSU President Dr. Virgilio Julius P. Manzano Jr., Extension Director Dr. Marlowe Aquino, and former MMSU President and PhilRice Executive Director Dr. Santiago R. Obien.

Source

- MMSU News, "MMSU President's Farm showcases 8 rice varieties," 4 November 2025 https://www.mmsu.edu.ph/article/news/2665/mmsu-presidents-farm-showcases-8-rice-varieties

Science High School Students Learn Sustainable Farming at MMSU's Demonstration Farm



PSHS-Ilocos Region Campus students at the MMSU Technology Demonstration Farm, Sarrat, 19 September 2025. (Photo: MMSU STRATCOM)

On September 19, 2025, 59 Grade 11 and 12 students of the Philippine Science High School–Ilocos Region Campus visited MMSU's Technology Demonstration Farm in Sta. Magdalena, Sarrat, as part of their community immersion program "The Science of Survival: A Community-Based Exploration," held during National Biotechnology Week. After an orientation at the ILAARRDEC One-Stop Information Shop at MMSU, Extension Director Dr. Marlowe Aquino discussed the role of biotechnology in agriculture and MMSU's hybrid off-season tomato varieties, and the students observed sustainable farming practices on site.

Source

- MMSU News, "PSHS-IRC students visit MMSU demo farm for sustainable farming practices," 3 October 2025 https://www.mmsu.edu.ph/news/pshs-irc-students-visit-mmsu-demo-farm-for-sustainable-farming-practices

MMSU Leads Soil Health Research for Rice-Based Cropping Systems



Kick-off meeting of the soil health management project with Griffith University and provincial agriculturists, 16 June 2025. (Photo: MMSU STRATCOM)

In 2025, MMSU became a key partner in a five-year research program led by Griffith University in Australia and funded by the Australian Centre for International Agricultural Research (ACIAR) and DOST-PCAARRD to develop soil knowledge, information and capacity for more productive and sustainable cropping systems in the Philippines (Project SLaM 2023/146). MMSU President Virgilio Julius P. Manzano Jr. pledged the university's commitment at the inception meeting held March 24–26, 2025 in Balanga City, Bataan.

MMSU leads the project on enhancing the productivity of rice-based cropping systems in Ilocos Norte, Tarlac and Bataan through soil health management. Led by Dr. Janet Villamor, it evaluates nutrient management practices and develops yield-calibrated soil health indices for rice-garlic, rice-rice and rice-onion systems. The kick-off meeting on June 16, 2025 at the MMSU Center for Flexible Learning brought together Griffith University, the country project manager and the provincial agriculturists of Ilocos Norte, Bataan and Tarlac, with visits to project sites in Batac, Vintar and Pasuquin.

Source

- MMSU News, "MMSU tapped as key partner in enhancing PH cropping systems through soil health management," 31 March 2025 https://www.mmsu.edu.ph/news/mmsu-tapped-as-key-partner-in-enhancing-ph-cropping-systems-through-soil-health-management
- MMSU News, "Griffith University, MMSU launch cropping system research in Luzon," 3 July 2025 https://www.mmsu.edu.ph/news/griffith-university-mmsu-launch-cropping-system-research-in-luzon

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.6 · Sustainable management of land for tourism (educational outreach)

Offer educational programme/outreach for local or national communities on sustainable management of land for tourism.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

DRAFT. The 2025 evidence is MMSU's rice paddy art, a farm-tourism attraction on university demonstration fields with student guides, and the TALON farm tourism project was approved in January 2026. Neither is yet a structured training for communities. Add visitor or tour numbers and any briefing given to visitors before submitting.

Evidence from 2025

Rice Paddy Art Brings Visitors to MMSU's Farm-Tourism Demonstration Fields



Unveiling of the 2025 rice paddy art featuring CHED Chairperson Shirley C. Agrupis and her ACHIEVE agenda, MMSU Extension Demonstration Area, 22 August 2025. (Photo: MMSU STRATCOM)



Rice paddy art featuring Dingras as the Rice Granary of Ilocos Norte, MMSU Dingras Campus, unveiled 9 September 2025. (Photo: MMSU STRATCOM)

In partnership with the Department of Agriculture–Philippine Rice Research Institute (DA-PhilRice), Mariano Marcos State University held the ceremonial transplanting of its eighth annual rice paddy art on July 7, 2025, at the Extension Demonstration Area near the Twin Gate of the Batac campus, and unveiled it on August 22, 2025. The design, featuring CHED Chairperson Shirley C. Agrupis and her ACHIEVE agenda, was realized with the PalayCheck System, the

Carrageenan Plant Growth Regulator, and two rice varieties, purple rice (IR 1552) and green rice (NSIC Rc222). About 50 BS Agriculture students, with MMSU extension and research staff and PhilRice, served as guides and transplanters. MMSU President Virgilio Julius P. Manzano Jr. said the paddy art "attracts visitors, promotes research, and showcases innovation," and Extension Director Dr. Marilou Lucas described it as a statement of commitment to agri-tourism, research and rice appreciation.

On September 9, 2025, the MMSU Dingras Campus unveiled its second rice paddy art, featuring Dingras as the Rice Granary of Ilocos Norte, created by MMSU student planters with PhilRice Ilocos Norte and the Local Government Unit of Dingras. Initiated in 2024, the Dingras paddy art is aligned with Republic Act 10816, the Farm Tourism Development Act, and CAFSD Dean Dr. Charlie B. Batin commended it for boosting agro-tourism and enrolment in agriculture programmes.

Source

- MMSU News, "CHED Chairperson Agrupis to be featured in MMSU-PhilRice paddy art," 8 July 2025 https://www.mmsu.edu.ph/article/news/2472/ched-chairperson-agrupis-to-be-featured-in-mmsu-philrice-paddy-art
- MMSU News, "2025 Rice Paddy art features CHED Chair Agrupis," 22 August 2025 https://www.mmsu.edu.ph/article/news/2527/2025-rice-paddy-art-features-ched-chair-agrupis
- MMSU News, "Dingras campus launches rice paddy art," 12 September 2025 https://www.mmsu.edu.ph/article/news/2560/dingras-campus-launches-rice-paddy-art

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

DOST-PCAARRD and UP-ISSI Assess MMSU's Proposed TALON Farm Tourism Project

Continuing implementation · site visit of January 15, 2026



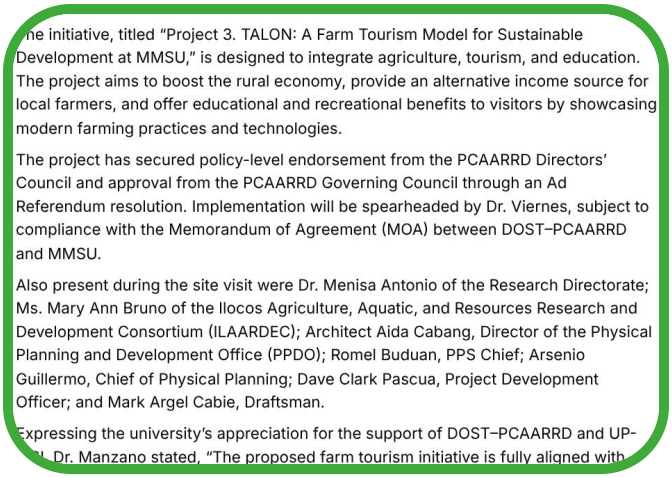
DOST-PCAARRD, UP-ISSI visit proposed MMSU farm tourism project site

By Wimbledon Bosque

Officials from the Department of Science and Technology–Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development (DOST–PCAARRD) and the University of the Philippines–Institute for Small-Scale Industries (UP-ISSI) conducted a site visit at the Mariano Marcos State University (MMSU) Crops Research Laboratory on January 15 to assess the proposed farm tourism project under the Science for the Convergence of Agriculture and Tourism (SciCAT) Expansion Program.

The delegation, led by Assistant Professor Melanie Moraga-Leaño, Director of UP-ISSI and SciCAT Program Leader, together with UP-ISSI extension specialists Kaye Bravo-Coballes, Janelle Faye Tanudtanud, Joshua Garcia, and Garry Nipales, along with Dianne Cabello of the Technology Transfer and Promotion Division, DOST–PCAARRD, [illegible] welcomed by MMSU President Virgilio Julius Manzano, Jr.

Capture: MMSU News, 20 January 2026, "DOST-PCAARRD, UP-ISSI visit proposed MMSU farm tourism project site" (headline and opening paragraphs).



[illegible]he initiative, titled "Project 3. TALON: A Farm Tourism Model for Sustainable Development at MMSU," is designed to integrate agriculture, tourism, and education. The project aims to boost the rural economy, provide an alternative income source for local farmers, and offer educational and recreational benefits to visitors by showcasing modern farming practices and technologies.

The project has secured policy-level endorsement from the PCAARRD Directors' Council and approval from the PCAARRD Governing Council through an Ad Referendum resolution. Implementation will be spearheaded by Dr. Viernes, subject to compliance with the Memorandum of Agreement (MOA) between DOST–PCAARRD and MMSU.

Also present during the site visit were Dr. Menisa Antonio of the Research Directorate; Ms. Mary Ann Bruno of the Ilocos Agriculture, Aquatic, and Resources Research and Development Consortium (ILAARDEC); Architect Aida Cabang, Director of the Physical Planning and Development Office (PPDO); Romel Buduan, PPS Chief; Arsenio Guillermo, Chief of Physical Planning; Dave Clark Pascua, Project Development Officer; and Mark Argel Cabie, Draftsman.

Expressing the university's appreciation for the support of DOST–PCAARRD and UP-[illegible]. Dr. Manzano stated, "The proposed farm tourism initiative is fully aligned with

Capture: the same article, on Project 3 TALON and the approval of the PCAARRD Governing Council.

On January 15, 2026, officials of the Department of Science and Technology–Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development (DOST-PCAARRD) and the University of the Philippines–Institute for Small-Scale Industries (UP-ISSI) visited the MMSU Crops Research Laboratory to assess a proposed farm tourism project under the Science for the Convergence of Agriculture and Tourism (SciCAT) Expansion Program. The delegation, led by UP-ISSI Director and SciCAT Program Leader Asst. Prof. Melanie Moraga-Leaño, was welcomed by MMSU President Virgilio Julius P. Manzano Jr. and guided through an on-site technical walkthrough by MMSU's Vice President for Research, Development and Innovation, the Director for Research, Innovation and Technology, the dean of the College of Business, Economics and Accountancy, and project members from its Department of Tourism and Hospitality Management led by Dr. Eva Jean Viernes. Planners of the Physical Planning and Development Office and the Ilocos Agriculture, Aquatic and Natural Resources Research and Development Consortium (ILAARRDEC) also took part.

The project, "Project 3. TALON: A Farm Tourism Model for Sustainable Development at MMSU," is designed to integrate agriculture, tourism and education. It aims to boost the rural economy, provide an alternative income source for local farmers, and offer educational and recreational benefits to visitors by showcasing modern farming practices and technologies. The project has the endorsement of the PCAARRD Directors' Council and the approval of the PCAARRD Governing Council through an Ad Referendum resolution, and its implementation, led by Dr. Viernes, is subject to the memorandum of agreement between DOST-PCAARRD and MMSU. MMSU is among

the select institutions developing science-and-technology-based agri-tourism hubs under SciCAT, which turns farms into tourism sites to promote innovation, entrepreneurship and community development.

Source

- MMSU News, “DOST-PCAARRD, UP-ISSI visit proposed MMSU farm tourism project site,” 20 January 2026
  https://www.mmsu.edu.ph/article/news/2869/dost-pcaarrd-up-issi-visit-proposed-mmsu-farm-tourism-project-site

Biodiversity Conservation Learning Center and Leisure Park Is Enriched with 28 Native Tree Species

Continuing implementation · since 2023

On January 13, 2023, MMSU planted 120 more seedlings of 28 native tree species and two bamboo species at the Biodiversity Conservation Learning Center and Leisure Park on the Main Campus, led by then President Shirley C. Agrupis with administrators, faculty, staff and students. The park is used as a field laboratory for classes and was identified for development into a health and wellness tourism hub, a plan endorsed by the provincial government and aligned with the Health and Wellness Complex of the College of Medicine.

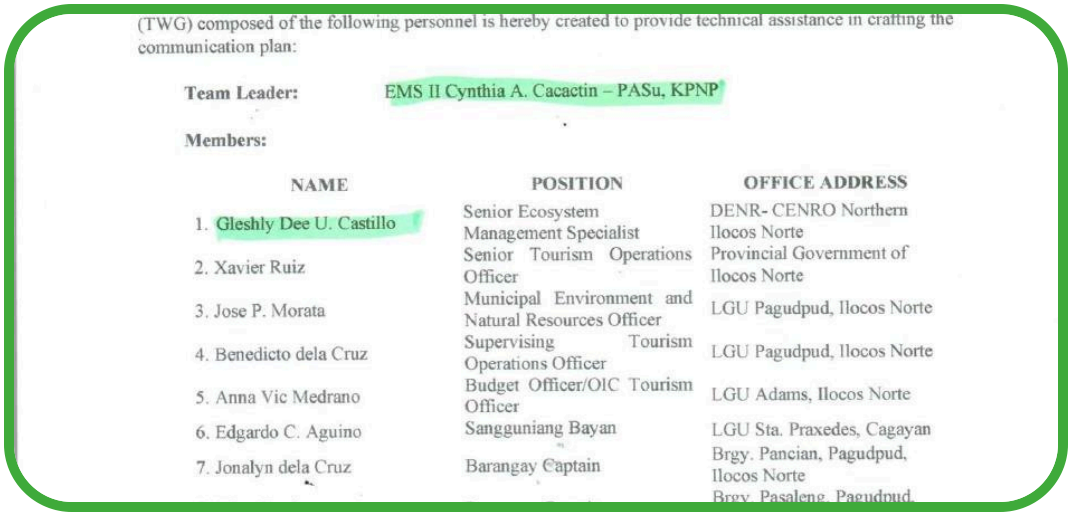
The planting continues a tradition that began in 2017 with the Green Wall project on the Batac campus and extends to the Nagbacsan site and the Currimao, Dingras and Laoag campuses, as the university’s contribution to the provincial “Green Wall of Ilocos Norte” and the National Greening Program of the Department of Environment and Natural Resources.

Source

- MMSU News, “More trees planted at MMSU to establish ‘health tourism’ hub,” 13 January 2023
  https://www.mmsu.edu.ph/news/more-trees-planted-at-mmsu-to-establish-health-tourism-hub

MMSU Expert Serves on the Kalbario-Patapat Natural Park Technical Working Group

Continuing role · appointment of 29 March 2023



(TWG) composed of the following personnel is hereby created to provide technical assistance in crafting the communication plan:

Team Leader: EMS II Cynthia A. Cacactin – PASu, KPNP

Members:

NAME	POSITION	OFFICE ADDRESS
1. Gleshly Dee U. Castillo	Senior Ecosystem Management Specialist	DENR- CENRO Northern Ilocos Norte
2. Xavier Ruiz	Senior Tourism Operations Officer	Provincial Government of Ilocos Norte
3. Jose P. Morata	Municipal Environment and Natural Resources Officer	LGU Pagudpud, Ilocos Norte
4. Benedicto dela Cruz	Supervising Tourism Operations Officer	LGU Pagudpud, Ilocos Norte
5. Anna Vic Medrano	Budget Officer/OIC Tourism Officer	LGU Adams, Ilocos Norte
6. Edgardo C. Aguino	Sangguniang Bayan	LGU Sta. Praxedes, Cagayan
7. Jonalyn dela Cruz	Barangay Captain	Brgy. Pancian, Pagudpud, Ilocos Norte
		Brgy. Pasaleng, Pagudpud,

Document: DENR-PENRO Ilocos Norte Special Order No. 2023-024 (29 March 2023) naming Dr. Arlene L. Gonzales of MMSU to the KPNP Technical Working Group.

Through DENR-PENRO Ilocos Norte Special Order No. 2023-024, dated March 29, 2023, Dr. Arlene L. Gonzales of MMSU was named a member of the Technical Working Group that prepares the Communication Plan of the Kalbario-Patapat Natural Park (KPNP), a protected area in Ilocos Norte. The group, led by the KPNP Protected Area Superintendent, brings together DENR offices, the Provincial Government of Ilocos Norte, the LGUs of Pagudpud and Adams in Ilocos Norte and Sta. Praxedes in Cagayan, barangay officials and Northwestern University.

The Communication Plan supports the goal of the KPNP Protected Area Management Plan to adopt and carry out effective information and education campaigns for biological diversity conservation and for the promotion of the protected area as a major ecotourism destination.

Source

- DENR-PENRO Ilocos Norte Special Order No. 2023-024, 29 March 2023 (MMSU SDG 15 page) https://www.mmsu.edu.ph/resources/1TCpM-DxbkgfOh4QGwlfVr8QavCchKWKu/data



MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.3.1 · Sustainable use, conservation and restoration of land (policy)

Have a policy to ensure the conservation, restoration and sustainable use of terrestrial ecosystems associated with the university, in particular forests, mountains and drylands.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

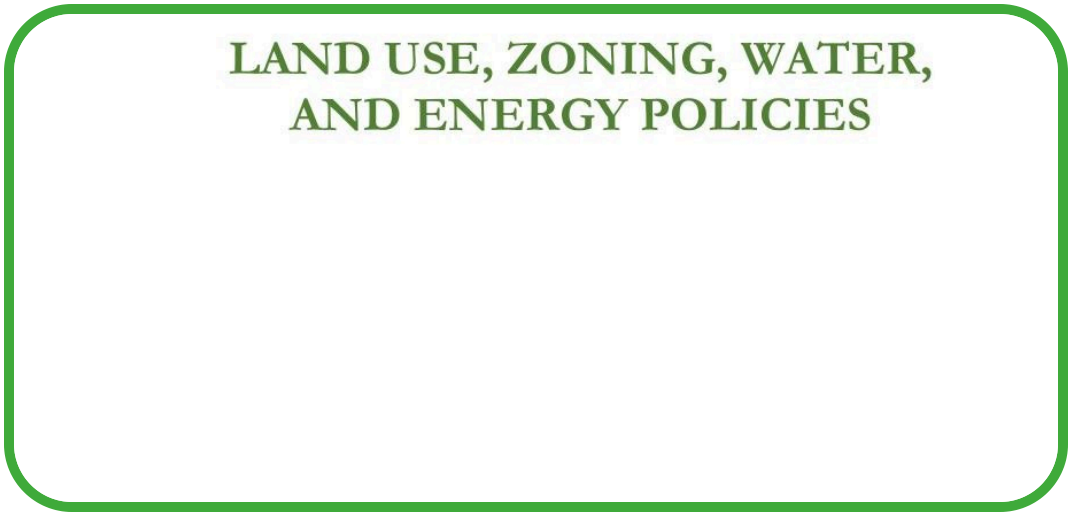
No 2025 item was found for this indicator. The material below is from before 2025 or from 2026 and, as far as the records show, is still in place.

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

MMSU Land Use Policy Commits the University to Conserve and Restore Its Lands

Policy · Sirmata 2040 LUDIP (2021)


LAND USE, ZONING, WATER,
AND ENERGY POLICIES

Document: MMSU Land Use, Zoning, Water and Energy Policies (Sirmata 2040 LUDIP, 2021), Land Use Policy, page 1.

MMSU's Land Use, Zoning, Water and Energy Policies, adopted as part of Sirmata 2040, the university's Land Use Development and Infrastructure Plan prepared in 2021 under Republic Act 11396, govern the conservation, restoration and sustainable use of all university lands across its campuses.

Under Policy 1.3, the allocation and use of university lands “shall aim to maintain, protect, conserve, restore, and enhance the integrity of the land itself, the environment and ecological balance in all its campuses.” Policy 7.1 requires decisions to be based on valid and reliable science, especially for any intent to alter or convert the natural state of any piece of university land, and Policy 10.2 gives fragile lands and protected areas special management “to protect, conserve, enrich, rehabilitate, and or sustain them.”

The zoning policy sets aside Protected Land as a distinct zone; under Policy 9.1, “protected landholdings of the university shall be conserved, preserved and to be excluded from any form of alteration except as may be permitted by operation of law.” The MMSU Forest Reserve is maintained as a no-build zone, and Policy 4.3 requires the university to regularly review, update and promptly communicate its land-use policies.

Source

- MMSU, Land Use, Zoning, Water and Energy Policies (Sirmata 2040 LUDIP, 2021) https://www.mmsu.edu.ph/resources/19uycXrgcO_iXPdPpMQwdCRxmoD1U5i60/view
- MMSU, Planning and Development on Local Biodiversity (SDG 15 evidence page) https://www.mmsu.edu.ph/resources/1tbtCaZa6Uh7ieu4epmz35VJ9jrRQGlPi/data



MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.3.3 · Local biodiversity included in planning and development

Include local biodiversity into any planning and development process (e.g. construction of new buildings).

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

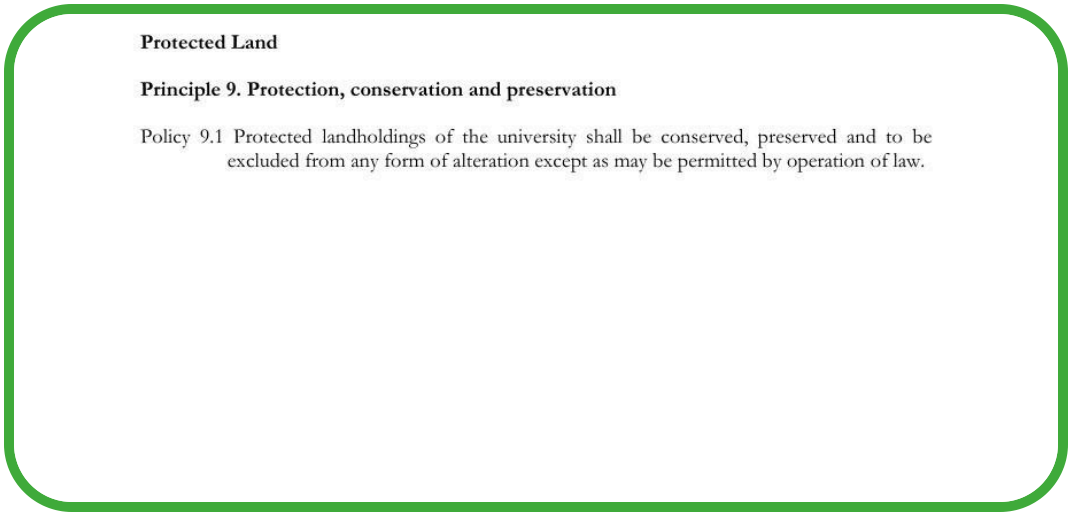
No 2025 item was found for this indicator. The material below is from before 2025 or from 2026 and, as far as the records show, is still in place.

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

Sirmata 2040: Planning MMSU's Campuses and Reserve Lands for Two Decades

Plan · Sirmata 2040 LUDIP (2021), a 20-year plan



Protected Land

Principle 9. Protection, conservation and preservation

Policy 9.1 Protected landholdings of the university shall be conserved, preserved and to be excluded from any form of alteration except as may be permitted by operation of law.

Document: MMSU Zoning Policy, Protected Land (Principle 9, Policy 9.1), Sirmata 2040 LUDIP.

Sirmata 2040, MMSU's Land Use Development and Infrastructure Plan (LUDIP), is the university's 20-year guide for managing its land resources and infrastructure. Prepared under Republic Act 11396, which requires state universities and colleges to have a LUDIP, and supported by a Php 2 million grant from the Commission on Higher Education (CHED), the plan covers the future land

use of the MMSU campuses and the university's reserve areas in Nagbacsan, Payao, Lubbot and Sarnap, City of Batac, including the forest reserves where MMSU's forestry and agroforestry research is set up.

Local biodiversity is built into the plan's development rules. The zoning plan sets aside Protected Land, which "shall be conserved, preserved and to be excluded from any form of alteration except as may be permitted by operation of law" (Policy 9.1), and the MMSU Forest Reserve is kept as a no-build zone. Before any land is altered or converted, the plan requires a science-based assessment of its natural suitability and carrying capacity (Policies 7.1, 10.1 and 11.1).

The plan was developed through stakeholder consultations, sectoral studies and campus planning workshops, and was endorsed by the CHED Technical Working Group after validation visits to the campuses.

Source

- MMSU, Land Use, Zoning, Water and Energy Policies (Sirmata 2040 LUDIP, 2021) https://www.mmsu.edu.ph/resources/19uycXrgcO_iXPdPpMQwdCRxmoD1U5i60/view
- MMSU, Planning and Development on Local Biodiversity (SDG 15 evidence page) https://www.mmsu.edu.ph/resources/1tbtCaZa6Uh7ieu4epmz35VJ9jrRQGlPi/data
- MMSU News, "CHED TWG OKs MMSU Sirmata 2040" https://www.mmsu.edu.ph/news/ched-twg-oks-mmsu-sirmata-2040
- MMSU News, "MMSU gets Php 2 M for LUDIP project" https://www.mmsu.edu.ph/news/mmsu-gets-php-2-m-for-ludip-project

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.3.5 · Collaboration for shared land ecosystems

Collaborate with the local community to maintain shared land ecosystems.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

Evidence from 2025

MMSU, TMP Foundation, DENR and Pasuquin LGU Sign Agreement to Conserve a Coastal Forest



Signing of the memorandum of agreement among MMSU, TMP Foundation, DENR and LGU-Pasuquin, 29 November 2025. (Photo: MMSU STRATCOM)



MMSU, TMP Foundation, DENR and LGU-Pasuquin officials at the agreement signing, 29 November 2025. (Photo: MMSU STRATCOM)

On November 29, 2025, Mariano Marcos State University, the Toyota Motor Philippines Foundation (TMPF), the Department of Environment and Natural Resources and the Local Government Unit of Pasuquin formalized through a memorandum of agreement a partnership to strengthen coastal forest conservation in Barangay Davila, Pasuquin, Ilocos Norte. The agreement was signed by MMSU President Virgilio Julius P. Manzano Jr. with TMPF leadership, Toyota Motor Philippines and Toyota Ilocos Norte executives, DENR-CENRO Bangui Ecosystem Management Specialist Justine Rose Teneza, and Sangguniang Bayan Member Romel Ranay for LGU-Pasuquin.

A mangrove planting followed the signing, with 1,750 propagules planted and another 1,750 planned for the following year. TMPF President Jose Maria Aligada described the partnership as shared stewardship to restore native mangrove species, rehabilitate coastal systems and boost

community resilience to climate-related hazards. Former MMSU College of Industrial Technology Dean Cesario Pacis was recognized as one of the initiators of the MMSU–Toyota partnership.

Source

- MMSU News, “MMSU ties up with Toyota and DENR for coastal preservation in Pasuquin town,” 4 December 2025 https://www.mmsu.edu.ph/news/mmsu-ties-up-with-toyota-and-denr-for-coastal-preservation-in-pasuquin-town
- The Manila Times, “TMP Foundation adopts mangrove forest for climate mitigation,” 10 December 2025 https://www.manilatimes.net/2025/12/10/tmt-newswire/tmp-foundation-adopts-mangrove-forest-for-climate-mitigation/2239598

MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand



MMSU, Rotary Club of Laoag Sunshine and LGU-Badoc plant mangrove propagules in Brgy. Pagsanahan, Badoc, 6 December 2025. (Photo: MMSU)



Participants of the mangrove enhancement and growing activity in Badoc, 6 December 2025. (Photo: MMSU)

On December 6, 2025, Mariano Marcos State University, the Rotary Club of Laoag Sunshine and the Local Government Unit of Badoc planted 280 mangrove propagules in Barangay Pagsanahan, Badoc, Ilocos Norte, to restore a mangrove stand extensively damaged by past typhoons and to ensure the continuity of ecological succession at the site.

The activity was led by MMSU President Virgilio Julius P. Manzano Jr., with the LGU of Badoc headed by Hon. Virgilio M. Calajate and the Rotary Club of Laoag Sunshine under Atty. Zsa Zsa Raval. MMSU’s Vice President for Resource Generation and its Directors for Extension, Curriculum Development and Innovative Learning, and Student Affairs and Development also took part.

Source

- MMSU News, “MMSU ties up with Rotary Club of Laoag Sunshine for mangrove planting in Badoc town,” 11 December 2025 https://www.mmsu.edu.ph/news/mmsu-ties-up-with-rotary-club-of-laoag-sunshine-for-mangrove-planting-in-badoc-town

MMSU Plants 250 Mangrove Propagules in Currimao on World Environment Day



Mangrove planting for World Environment Day, Brgy. Maglaoi, Currimao, June 2025. (Photo: MMSU STRATCOM)



MMSU, LGU Currimao, DENR and fisherfolk volunteers at the planting site, June 2025. (Photo: MMSU STRATCOM)

In June 2025, in celebration of World Environment Day, Mariano Marcos State University planted 250 mangrove propagules along the coastal area of Barangay Maglaoi, Currimao, Ilocos Norte, to restore degraded shorelines, enhance climate resilience and promote sustainable coastal resource management. The activity was spearheaded by the MMSU Sustainability and Futures Thinking (SFT) Office with the Local Government Unit of Currimao and led by Dr. Mee Jay Domingo, Vice President for Planning and Strategic Foresight.

MMSU faculty, education support personnel and students joined representatives of the Department of Environment and Natural Resources, barangay officials, members of local fisherfolk organizations and community members. SFT Director Dr. Arlene Gonzales explained that mangroves were chosen because they act as natural filters that intercept plastic waste from land before it reaches the sea, while absorbing carbon, protecting shorelines from erosion and providing habitat for marine life.

Source

- MMSU News, “MMSU plants 250 mangrove propagules in Currimao,” 11 June 2025 https://www.mmsu.edu.ph/news/mmsu-plants-250-mangrove-propagules-in-currimao

MMSU and DENR Region 1 Strengthen Partnership for Mangrove Conservation



Dr. Arlene L. Gonzales at the MMSU–DENR Region 1 dialogue on mangrove conservation, ILAARRDEC.



MMSU and DENR Region 1 officials during the dialogue at ILAARRDEC.

MMSU, in partnership with the Department of Environment and Natural Resources (DENR) Region 1, convened a dialogue at ILAARRDEC to advance mangrove conservation and coastal resilience in the Ilocos Region through science-based and community-centered approaches. Central to the dialogue was the DOST-PCAARRD-funded nature-based solutions mangrove project led by Dr. Arlene Gonzales.

DENR officials pledged technical support and active participation in restoration, monitoring and evaluation activities. The same project brought representatives of Currimao, Pasuquin, Badoc and Burgos, with DENR-PENRO, to MMSU's mangrove propagation seminar-workshop on October 27–28, 2025.

Source

- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking
- MMSU News, "MMSU trains Ilocos Norte coastal towns on mangrove propagation and management," 5 November 2025 https://www.mmsu.edu.ph/article/news/2670/mmsu-trains-ilocos-norte-coastal-towns-on-mangrove-propagation-and-management

Indigenous Food Plant Foodscapes Established with Vintar and San Nicolas Partners



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

In October 2025, the MMSU Extension and Research Directorates partnered with local government units and schools to establish Community Foodscapes and Seedbanks for Indigenous Food Plants. Held at Eladio V. Barangan Memorial Elementary School in San Nicolas, Ilocos Norte, the activity brought together officials of Barangays San Roque and Parut, Vintar, the Municipal Agriculture Office of Vintar, agricultural extension workers, school leaders and community representatives as the second batch of partner sites.

The partner communities maintain the foodscapes as shared seedbanks and learning gardens, with planting materials and technical assistance from MMSU, to conserve indigenous food resources such as kapaskapas, alokon, pako, dalayap, wild singkamas and roselle. They join existing foodscape sites in Sarrat and Batac.

Source

- MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025 https://extension.mmsu.edu.ph/news/show/389/mmsu-expands-indigenous-food-plant-seedbanks-and-foodscapes-in-lgus
- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

A New Bambusetum in Barangay Aglipay, Pinili



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

During a two-day bamboo production training in 2025, the MMSU Extension Directorate and CAFSD, with Barangay Aglipay, Pinili, and the support of ILAARRDEC, worked with 46 farmers, local leaders and community members to establish a new bambusetum, a collection of bamboo species, in Barangay Aglipay. The training covered bamboo propagation, nursery establishment and plantation management.

Source

- MMSU News, “Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili,” 3 September 2025 https://www.mmsu.edu.ph/news/sustainable-livelihood-mmsu-ilaarrdec-hold-bamboo-training-in-pinili

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

Mangrove Nature-based Solutions Project Runs from 2024 to 2026

Continuing implementation · 1 May 2024 to 30 April 2026



Project Leader Dr. Arlene L. Gonzales presents the project at its launch, MMSU Review Center, 27 June 2024. (Image credit: TTPD, DOST-PCAARRD)



The MMSU project team with DOST-PCAARRD, DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the LGUs of Badoc and Currimao at the launch. (Image credit: TTPD, DOST-PCAARRD)

The DOST-PCAARRD-funded project "Nature-based Solutions (NbS) for Sustainable Mangrove Management and Resilient Coastal Communities Through Innovative Green Engineering Approach," led by Dr. Arlene L. Gonzales of MMSU, was launched at the MMSU Review Center on June 27, 2024, and runs from May 1, 2024 to April 30, 2026 in Badoc and Currimao, Ilocos Norte. DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the local government units of Badoc and Currimao support the project.

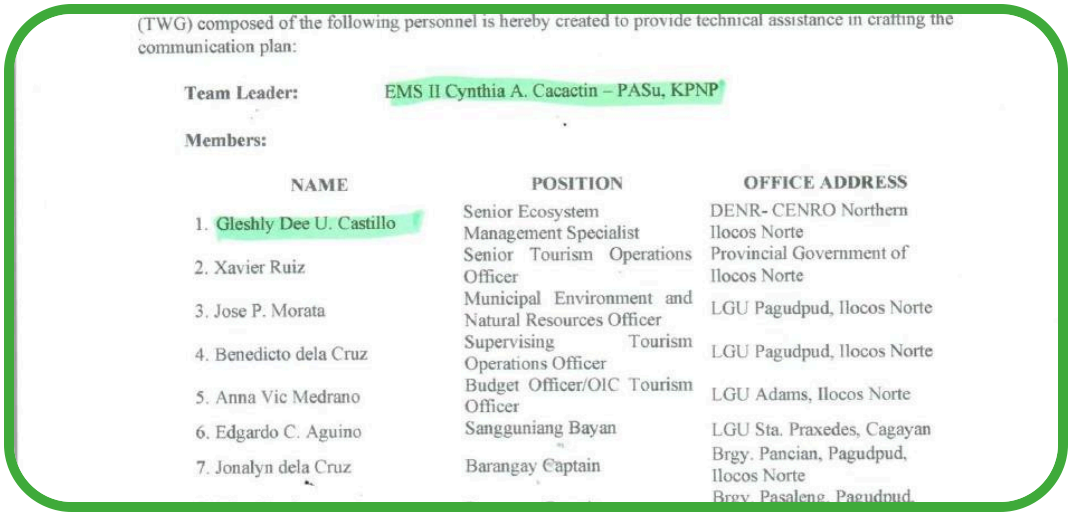
The 2025 activities of the project, including the dialogue with DENR Region 1 and the October 2025 mangrove propagation seminar-workshop, are described under Indicators 15.2.3 and 15.2.4.

Source

- DOST-PCAARRD, "DOST-PCAARRD, MMSU launch nature-based solutions project on mangrove management in Ilocos Norte," 7 August 2024
  https://www.pcaarrd.dost.gov.ph/index.php/quick-information-dispatch-qid-articles/dost-pcaarrd-mmsu-launch-nature-based-solutions-project-on-mangrove-management-in-ilocos-norte

MMSU Expert Serves on the Kalbario-Patapat Natural Park Technical Working Group

Continuing role · appointment of 29 March 2023



(TWG) composed of the following personnel is hereby created to provide technical assistance in crafting the communication plan:

Team Leader: EMS II Cynthia A. Cacactin – PASu, KPNP

Members:

NAME	POSITION	OFFICE ADDRESS
1. Gleshly Dee U. Castillo	Senior Ecosystem Management Specialist	DENR- CENRO Northern Ilocos Norte
2. Xavier Ruiz	Senior Tourism Operations Officer	Provincial Government of Ilocos Norte
3. Jose P. Morata	Municipal Environment and Natural Resources Officer	LGU Pagudpud, Ilocos Norte
4. Benedicto dela Cruz	Supervising Tourism Operations Officer	LGU Pagudpud, Ilocos Norte
5. Anna Vic Medrano	Budget Officer/OIC Tourism Officer	LGU Adams, Ilocos Norte
6. Edgardo C. Aguino	Sangguniang Bayan	LGU Sta. Praxedes, Cagayan
7. Jonalyn dela Cruz	Barangay Captain	Brgy. Pancian, Pagudpud, Ilocos Norte
		Brgy. Pasaleng, Pagudpud,

Document: DENR-PENRO Ilocos Norte Special Order No. 2023-024 (29 March 2023) naming Dr. Arlene L. Gonzales of MMSU to the KPNP Technical Working Group.

Through DENR-PENRO Ilocos Norte Special Order No. 2023-024, dated March 29, 2023, Dr. Arlene L. Gonzales of MMSU was named a member of the Technical Working Group that prepares the Communication Plan of the Kalbario-Patapat Natural Park (KPNP), a protected area in Ilocos Norte. The group, led by the KPNP Protected Area Superintendent, brings together DENR offices, the Provincial Government of Ilocos Norte, the LGUs of Pagudpud and Adams in Ilocos Norte and Sta. Praxedes in Cagayan, barangay officials and Northwestern University.

The Communication Plan supports the goal of the KPNP Protected Area Management Plan to adopt and carry out effective information and education campaigns for biological diversity conservation and for the promotion of the protected area as a major ecotourism destination.

Source

- DENR-PENRO Ilocos Norte Special Order No. 2023-024, 29 March 2023 (MMSU SDG 15 page) https://www.mmsu.edu.ph/resources/1TCpM-DxbkgfOh4QGwlfVr8QavCchKWKu/data

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

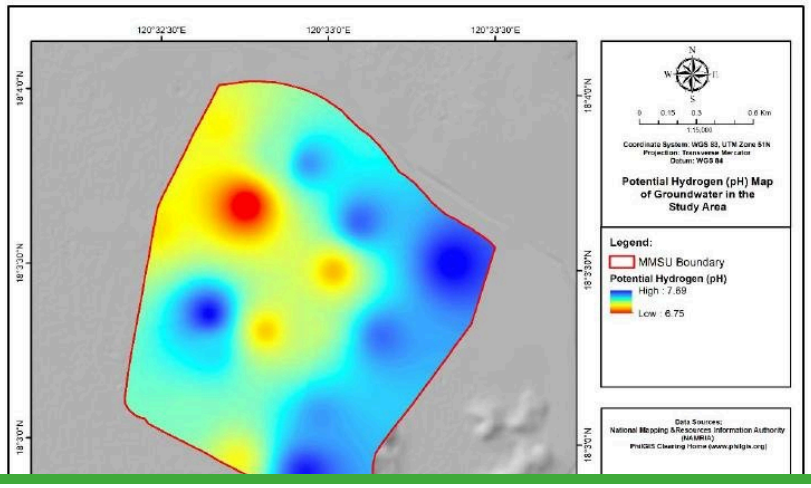
SDG 15.4.1 · Water discharge guidelines and standards

Have water quality standards and guidelines for water discharges (to uphold water quality in order to protect ecosystems, wildlife, and human health and welfare).

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

Evidence from 2025

Groundwater Quality Monitoring at the MMSU Farm



Document: Groundwater Quality Assessment of the MMSU Farm, spatial variation of pH (monitoring through June 2025).

Through June 2025, MMSU continued monthly monitoring of 17 groundwater wells on the MMSU Farm, measuring pH, total dissolved solids, electrical conductivity, chloride and nitrate-nitrogen and comparing the results with the Philippine National Standards for Drinking Water (2017). The assessment, covering February 2024 to June 2025, flags wells with persistently high dissolved solids, which rose again from January to May 2025, for continued monitoring to guide water resource management and prevent further degradation of groundwater quality.

Source

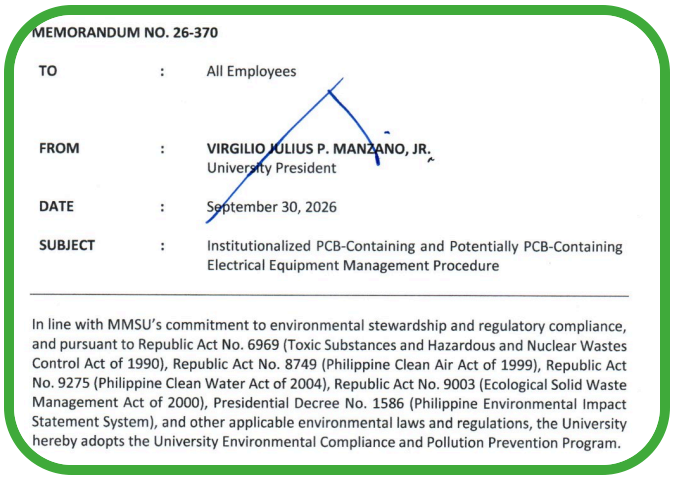
- MMSU, Groundwater Quality Assessment of the MMSU Farm (monitoring February 2024 – June 2025) https://www.mmsu.edu.ph/resources/15g9Xtys7q3Rujd0-CZvPH4OHTdJ1__Z8/data

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

Memorandum No. 26-370 Institutionalizes MMSU's Environmental Compliance and Hazardous Waste Procedures

Continuing implementation · 30 September 2026



MEMORANDUM NO. 26-370

TO : All Employees

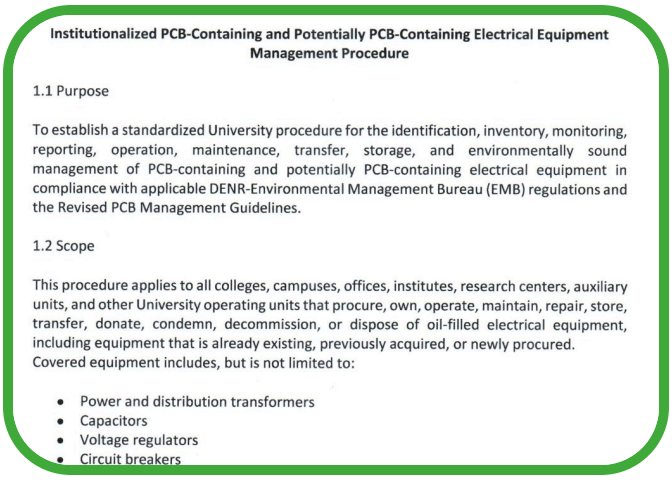
FROM : VIRGILIO JULIUS P. MANZANO, JR.
University President

DATE : September 30, 2026

SUBJECT : Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure

In line with MMSU's commitment to environmental stewardship and regulatory compliance, and pursuant to Republic Act No. 6969 (Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990), Republic Act No. 8749 (Philippine Clean Air Act of 1999), Republic Act No. 9275 (Philippine Clean Water Act of 2004), Republic Act No. 9003 (Ecological Solid Waste Management Act of 2000), Presidential Decree No. 1586 (Philippine Environmental Impact Statement System), and other applicable environmental laws and regulations, the University hereby adopts the University Environmental Compliance and Pollution Prevention Program.

Document: Office of the President Memorandum No. 26-370, 30 September 2026, adopting the University Environmental Compliance and Pollution Prevention Program.



Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure

1.1 Purpose

To establish a standardized University procedure for the identification, inventory, monitoring, reporting, operation, maintenance, transfer, storage, and environmentally sound management of PCB-containing and potentially PCB-containing electrical equipment in compliance with applicable DENR-Environmental Management Bureau (EMB) regulations and the Revised PCB Management Guidelines.

1.2 Scope

This procedure applies to all colleges, campuses, offices, institutes, research centers, auxiliary units, and other University operating units that procure, own, operate, maintain, repair, store, transfer, donate, condemn, decommission, or dispose of oil-filled electrical equipment, including equipment that is already existing, previously acquired, or newly procured.
Covered equipment includes, but is not limited to:

- Power and distribution transformers
- Capacitors
- Voltage regulators
- Circuit breakers

Document: attached procedure for PCB-containing and potentially PCB-containing electrical equipment (purpose, scope and inventory), Memorandum No. 26-370.

On September 30, 2026, University President Virgilio Julius P. Manzano Jr. issued Memorandum No. 26-370 to all employees, adopting the University Environmental Compliance and Pollution Prevention Program under Republic Act 6969 (Toxic Substances and Hazardous and Nuclear Wastes Control Act), Republic Act 8749 (Clean Air Act), Republic Act 9275 (Clean Water Act), Republic Act 9003 (Ecological Solid Waste Management Act) and Presidential Decree 1586 (Environmental Impact Statement System). The memorandum covers the management of hazardous wastes, wastewater, air emissions, solid wastes and chemicals, together with permits, monitoring, reporting and inspections required by the Department of Environment and Natural Resources – Environmental Management Bureau (DENR-EMB), and directs all offices, colleges, laboratories, research units and auxiliary enterprises to comply with the attached Environmental Compliance Procedures.

The attached procedure for PCB-containing and potentially PCB-containing electrical equipment, such as transformers, capacitors, voltage regulators, circuit breakers, reclosers and oil-filled switches, requires every unit to inventory its equipment and submit the inventory to the Sustainability and Futures Thinking (SFT) Office. The SFT Office validates the inventory, keeps the university's central PCB inventory and the DENR-EMB online PCB inventory, ensures labelling, conducts inspections and annual inventory validation, and prepares the reports required by DENR-EMB. No covered equipment may be relocated, transferred, repaired through oil handling,

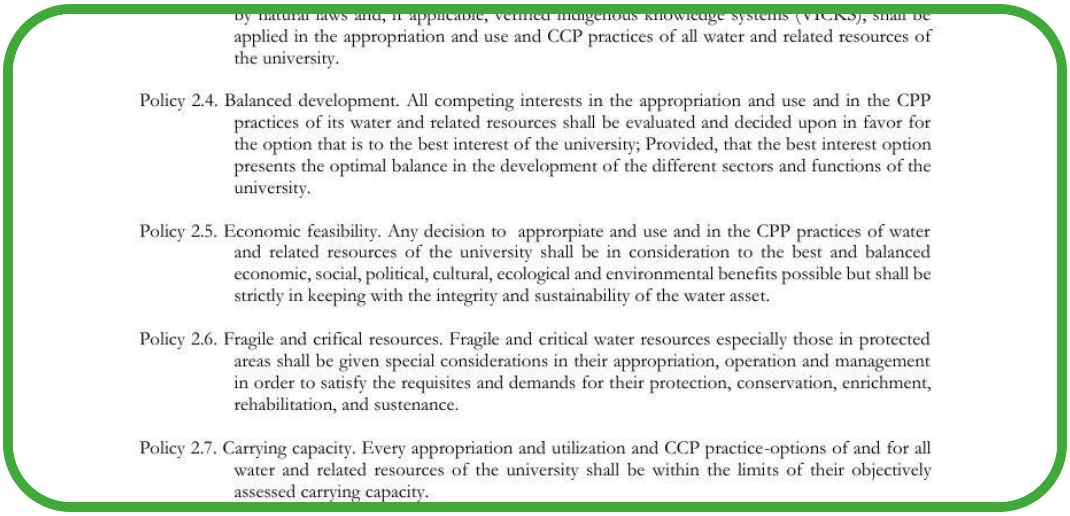
decommissioned, condemned or disposed of without the SFT Office's written clearance, and PCB wastes and equipment may be disposed of only through DENR-accredited transporters, treaters and disposal facilities. The SFT Office keeps hazardous waste manifests and certificates of treatment or disposal among its records.

Source

- MMSU Office of the President, Memorandum No. 26-370, "Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure," 30 September 2026

MMSU Water Policy Protects University Waters from Pollution

Policy · Sirmata 2040 (2021) and Integrated Pollution and Waste Management Program (revised 2020)



by natural laws and, if applicable, verified indigenous knowledge systems (VIKS), shall be applied in the appropriation and use and CCP practices of all water and related resources of the university.

Policy 2.4. Balanced development. All competing interests in the appropriation and use and in the CPP practices of its water and related resources shall be evaluated and decided upon in favor for the option that is to the best interest of the university; Provided, that the best interest option presents the optimal balance in the development of the different sectors and functions of the university.

Policy 2.5. Economic feasibility. Any decision to approrpiate and use and in the CPP practices of water and related resources of the university shall be in consideration to the best and balanced economic, social, political, cultural, ecological and environmental benefits possible but shall be strictly in keeping with the integrity and sustainability of the water asset.

Policy 2.6. Fragile and crifical resources. Fragile and critical water resources especially those in protected areas shall be given special considerations in their appropriation, operation and management in order to satisfy the requisites and demands for their protection, conservation, enrichment, rehabilitation, and sustenance.

Policy 2.7. Carrying capacity. Every appropriation and utilization and CCP practice-options of and for all water and related resources of the university shall be within the limits of their objectively assessed carrying capacity.

Document: MMSU Water Policy, control, conservation and protection of waters (Policies 2.6–2.9), Sirmata 2040 LUDIP.

MMSU's Water Policy under Sirmata 2040, based on the Water Code of the Philippines and its 2005 implementing rules, requires that "all water and related resources of the university, ground and surface, shall be shielded from any kind of pollution" (Policy 2.9) and that the rehabilitation of polluted resources be immediate, giving importance to water impoundments, catch basins and water treatment facilities (Policy 2.8).

The university's Integrated Pollution and Waste Management Program also provides for registration as a hazardous waste generator for wastewater discharge from agro-industrial facilities such as the piggery, in compliance with Republic Act 9275, the Philippine Clean Water Act.

Source

- MMSU, Land Use, Zoning, Water and Energy Policies (Sirmata 2040 LUDIP, 2021) https://www.mmsu.edu.ph/resources/19uycXrgcO_iXPdPpMQwdCRxmoD1U5i60/view
- MMSU, Revised Integrated Pollution and Waste Management Program https://www.mmsu.edu.ph/resources/1fNUCZwyOozOKO27L2Bm6pNN-P_cJNpLd/view
- MMSU, Water Discharge Management (SDG 15 evidence page) https://www.mmsu.edu.ph/resources/1l8foDfenNc7xpKp8wM8XShfZT58YyjSH/data



MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.4.2 · Policy on plastic waste reduction

Have a policy on reducing plastic waste on campus.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

Evidence from 2025

MMSU Commits to the DENR-EMB Region I Regional Ecology Center

On July 1, 2025, in Bauang, La Union, MMSU signed a memorandum of commitment with the DENR–Environmental Management Bureau Region I and other government agencies, private sector representatives, non-government organizations and academic institutions for the Regional Ecology Center. MMSU was represented by Vice President for Planning and Strategic Foresight Mee Jay Domingo and Campus Sustainability Chief Ronnel Pagurayan.

Under the commitment, MMSU supports improved solid waste management under Republic Act 9003 (Ecological Solid Waste Management Act) and Republic Act 11898 (Extended Producer Responsibility Act) through education and public information campaigns and the development of informational materials.

Source

- MMSU News, "MMSU commits to ecological solid waste management in Region I," 2 July 2025 https://www.mmsu.edu.ph/news/mmsu-commits-to-ecological-solid-waste-management-in-region-i

Student Advocates Hold Monthly Clean-Up Drive on Campus



MMSU Advocates for Cultural Development clean-up drive on campus, 24 October 2025. (Photo: MMSU)

On October 24, 2025, the MMSU Advocates for Cultural Development (MACD), an organization representing students from indigenous peoples' communities, held its monthly clean-up drive, "iPaglilinis," around the university grounds, the Student Center and parts of the University Lagoon. About 30 bags of waste were collected and segregated into plastic, paper and tin materials.

Source

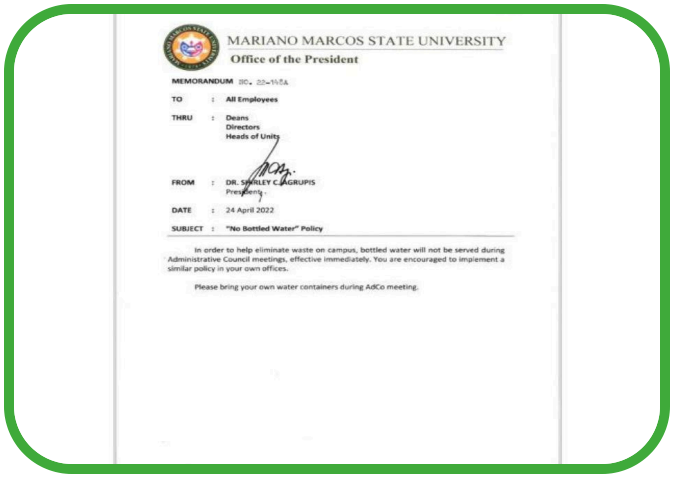
- MMSU News, "MACD leads clean-up drive in MMSU," 29 October 2025
  https://www.mmsu.edu.ph/news/macd-leads-clean-up-drive-in-mmsu

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

MMSU Policies Eliminate Styrofoam and Reduce Plastic Waste on Campus

Policies · 2009, 2015, 2016, 2022 and 2023, with 2024 and 2025 actions



MARIANO MARCOS STATE UNIVERSITY
Office of the President

MEMORANDUM NO. 22-148A

TO : All Employees

THRU : Deans
Directors
Heads of Units

FROM : DR. SHIRLEY C. AGRUPIS
President

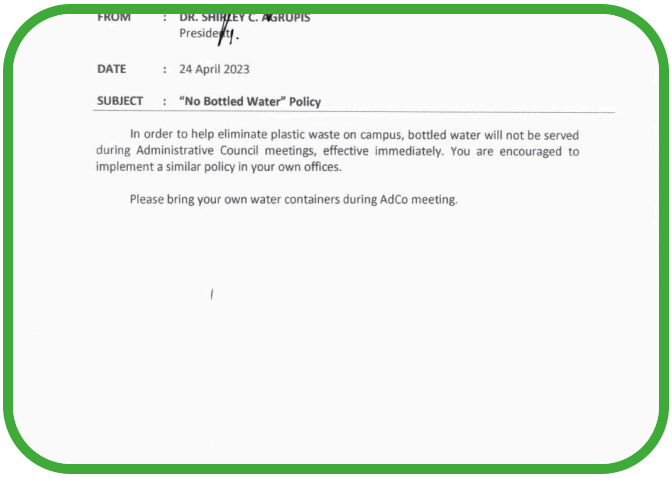
DATE : 24 April 2022

SUBJECT : "No Bottled Water" Policy

In order to help eliminate waste on campus, bottled water will not be served during Administrative Council meetings, effective immediately. You are encouraged to implement a similar policy in your own offices.

Please bring your own water containers during AdCo meeting.

Document: Office of the President Memorandum No. 22-148A, "No Bottled Water" Policy, 24 April 2022.



FROM : DR. SHIRLEY C. AGRUPIS
President

DATE : 24 April 2023

SUBJECT : "No Bottled Water" Policy

In order to help eliminate plastic waste on campus, bottled water will not be served during Administrative Council meetings, effective immediately. You are encouraged to implement a similar policy in your own offices.

Please bring your own water containers during AdCo meeting.

Document: Office of the President Memorandum No. 23-189, "No Bottled Water" Policy, 24 April 2023.

MMSU reduces plastic waste under its Integrated Waste Management Program, approved by the Board of Regents in 2009 and amended in 2015, and its revised Integrated Pollution and Waste Management Program. The program prohibits Styrofoam for serving or packing food in the university Food Service Unit and limits the use of plastics, and a policy promulgated on February 11, 2016 eliminates Styrofoam and plastic as secondary packaging materials in favour of paper bags, cartons, eco-bags and reusable containers. The University Waste Management Committee develops, reviews and monitors these rules, and the food service unit requires customers dining out to bring their own food containers.

The Office of the President reissues the "No Bottled Water" Policy each year: Memorandum No. 22-148A of April 24, 2022 and Memorandum No. 23-189 of April 24, 2023 stop the serving of bottled water at Administrative Council meetings, ask members to bring their own water containers, and encourage all offices to adopt the same policy. In October 2024 the College of Agriculture, Food and Sustainable Development banned plastic bottled water in its auxiliary

buildings, and on July 1, 2025 MMSU signed a memorandum of commitment with DENR-EMB Region I for the Regional Ecology Center to support solid waste management under Republic Act 9003 and the Extended Producer Responsibility Act (Republic Act 11898).

Source

- MMSU, Plastic Waste Management (SDG 15 evidence page) https://www.mmsu.edu.ph/resources/1NU-Qda9KG0J-PIOgKZwPKC4h_TU05zcm/data
- MMSU, Revised Integrated Pollution and Waste Management Program https://www.mmsu.edu.ph/resources/1fNUCZwyOozOKO27L2Bm6pNN-P_cJNpLd/view
- MMSU Office of the President, “No Bottled Water” Policy, Memorandum No. 22-148A (2022) and No. 23-189 (2023) https://www.mmsu.edu.ph/resources/1LbnK8yYkg-47ChbXyT0r8VQ9eBGuMxRx/data
- MMSU News, “MMSU CAFSD rolls out waste and pollution management program,” 15 October 2024 https://www.mmsu.edu.ph/news/mmsu-cafsd-rolls-out-waste-and-pollution-management-program
- MMSU News, “MMSU commits to ecological solid waste management in Region I,” 2 July 2025 https://www.mmsu.edu.ph/news/mmsu-commits-to-ecological-solid-waste-management-in-region-i



MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.4.3 · Policy on hazardous waste disposal

Have a policy, process or practice on waste disposal covering hazardous materials.

Evidence year: 2025 (calendar year), with continuing implementation 2024–2026

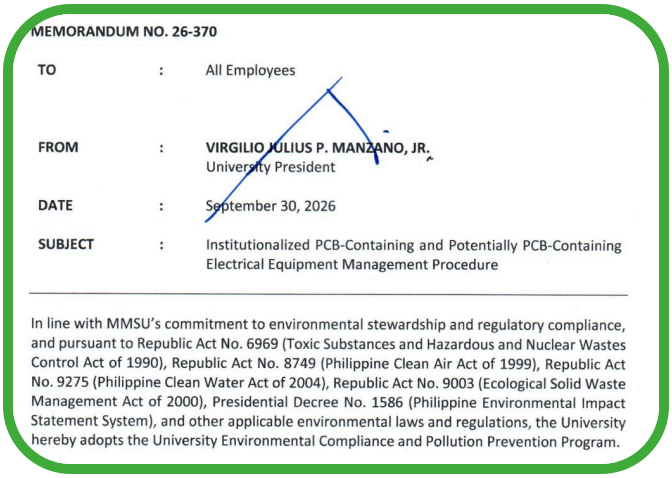
No 2025 item was found for this indicator. The material below is from before 2025 or from 2026 and, as far as the records show, is still in place.

Continuing implementation

Policies and programmes that began before 2025 and continue to be implemented, with 2024 and 2026 activities that show the programme is ongoing. THE accepts policies that are already in place.

Memorandum No. 26-370 Institutionalizes MMSU's Environmental Compliance and Hazardous Waste Procedures

Continuing implementation · 30 September 2026



MEMORANDUM NO. 26-370

TO : All Employees

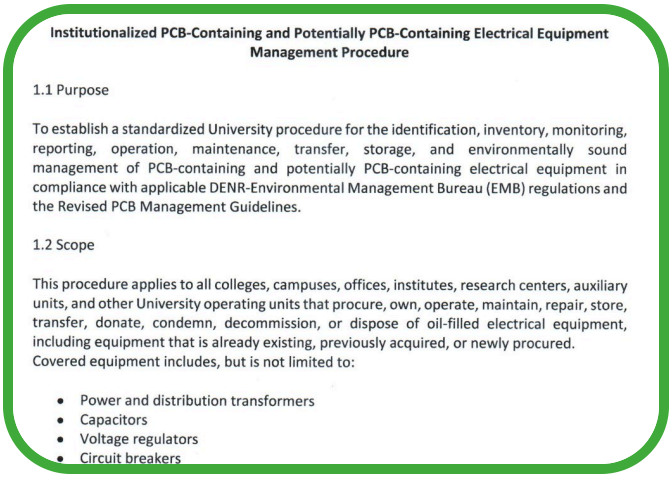
FROM : VIRGILIO JULIUS P. MANZANO, JR.
University President

DATE : September 30, 2026

SUBJECT : Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure

In line with MMSU's commitment to environmental stewardship and regulatory compliance, and pursuant to Republic Act No. 6969 (Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990), Republic Act No. 8749 (Philippine Clean Air Act of 1999), Republic Act No. 9275 (Philippine Clean Water Act of 2004), Republic Act No. 9003 (Ecological Solid Waste Management Act of 2000), Presidential Decree No. 1586 (Philippine Environmental Impact Statement System), and other applicable environmental laws and regulations, the University hereby adopts the University Environmental Compliance and Pollution Prevention Program.

Document: Office of the President Memorandum No. 26-370, 30 September 2026, adopting the University Environmental Compliance and Pollution Prevention Program.



Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure

1.1 Purpose

To establish a standardized University procedure for the identification, inventory, monitoring, reporting, operation, maintenance, transfer, storage, and environmentally sound management of PCB-containing and potentially PCB-containing electrical equipment in compliance with applicable DENR-Environmental Management Bureau (EMB) regulations and the Revised PCB Management Guidelines.

1.2 Scope

This procedure applies to all colleges, campuses, offices, institutes, research centers, auxiliary units, and other University operating units that procure, own, operate, maintain, repair, store, transfer, donate, condemn, decommission, or dispose of oil-filled electrical equipment, including equipment that is already existing, previously acquired, or newly procured.
Covered equipment includes, but is not limited to:

- Power and distribution transformers
- Capacitors
- Voltage regulators
- Circuit breakers

Document: attached procedure for PCB-containing and potentially PCB-containing electrical equipment (purpose, scope and inventory), Memorandum No. 26-370.

On September 30, 2026, University President Virgilio Julius P. Manzano Jr. issued Memorandum No. 26-370 to all employees, adopting the University Environmental Compliance and Pollution Prevention Program under Republic Act 6969 (Toxic Substances and Hazardous and Nuclear Wastes Control Act), Republic Act 8749 (Clean Air Act), Republic Act 9275 (Clean Water Act),

Republic Act 9003 (Ecological Solid Waste Management Act) and Presidential Decree 1586 (Environmental Impact Statement System). The memorandum covers the management of hazardous wastes, wastewater, air emissions, solid wastes and chemicals, together with permits, monitoring, reporting and inspections required by the Department of Environment and Natural Resources – Environmental Management Bureau (DENR-EMB), and directs all offices, colleges, laboratories, research units and auxiliary enterprises to comply with the attached Environmental Compliance Procedures.

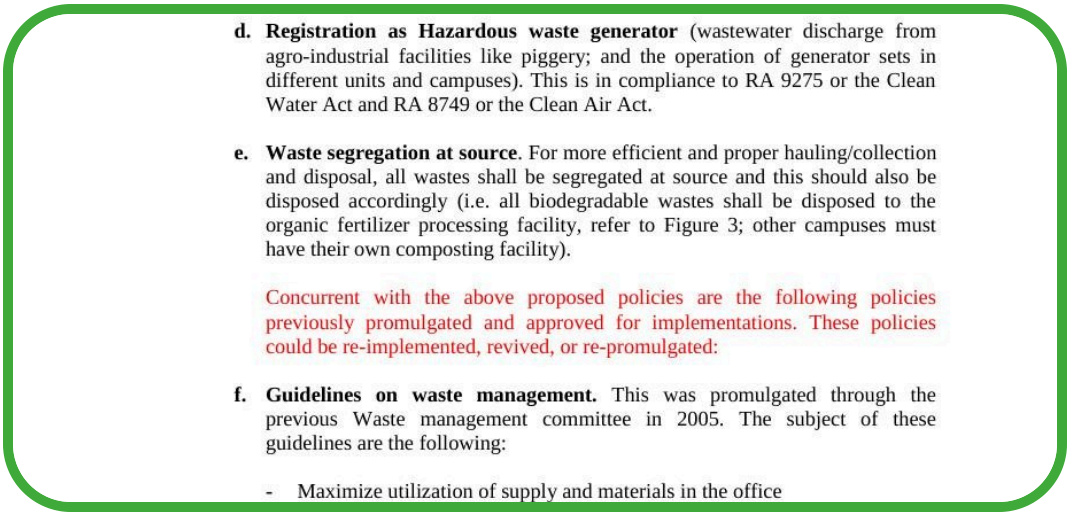
The attached procedure for PCB-containing and potentially PCB-containing electrical equipment, such as transformers, capacitors, voltage regulators, circuit breakers, reclosers and oil-filled switches, requires every unit to inventory its equipment and submit the inventory to the Sustainability and Futures Thinking (SFT) Office. The SFT Office validates the inventory, keeps the university's central PCB inventory and the DENR-EMB online PCB inventory, ensures labelling, conducts inspections and annual inventory validation, and prepares the reports required by DENR-EMB. No covered equipment may be relocated, transferred, repaired through oil handling, decommissioned, condemned or disposed of without the SFT Office's written clearance, and PCB wastes and equipment may be disposed of only through DENR-accredited transporters, treaters and disposal facilities. The SFT Office keeps hazardous waste manifests and certificates of treatment or disposal among its records.

Source

- MMSU Office of the President, Memorandum No. 26-370, "Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure," 30 September 2026

Hazardous Waste Disposal under MMSU's Integrated Pollution and Waste Management Program

Policy · Integrated Pollution and Waste Management Program (revised 2020)



d. **Registration as Hazardous waste generator** (wastewater discharge from agro-industrial facilities like piggery; and the operation of generator sets in different units and campuses). This is in compliance to RA 9275 or the Clean Water Act and RA 8749 or the Clean Air Act.

e. **Waste segregation at source**. For more efficient and proper hauling/collection and disposal, all wastes shall be segregated at source and this should also be disposed accordingly (i.e. all biodegradable wastes shall be disposed to the organic fertilizer processing facility, refer to Figure 3; other campuses must have their own composting facility).

Concurrent with the above proposed policies are the following policies previously promulgated and approved for implementations. These policies could be re-implemented, revived, or re-promulgated:

f. **Guidelines on waste management.** This was promulgated through the previous Waste management committee in 2005. The subject of these guidelines are the following:

- Maximize utilization of supply and materials in the office

Document: Revised Integrated Pollution and Waste Management Program of MMSU, provisions on PCB and hazardous waste storage and registration.

MMSU's Integrated Waste Management Program was revised in 2020 into the Integrated Pollution and Waste Management Program (IPWMP), implemented by the Pollution Control Unit and the General Services directorate, to cover hazardous materials. The IPWMP assigns focal persons for hazardous wastes (chemical and related wastes), air and water pollution, solid wastes, and PCBs and related industrial wastes, and requires the waste coordinators of every unit to store and label generated hazardous wastes properly before collection.

The program establishes centralized, separate storage facilities for polychlorinated biphenyls (PCBs) and other hazardous or toxic wastes collected from the campuses, where the wastes are kept for up to one year before disposal through a DENR-accredited transport, storage and disposal facility, in compliance with Republic Act 6969 and its storage guidelines. Units that generate hazardous waste are registered as hazardous waste generators, and the university coordinates with the Environmental Management Bureau of DENR for final collection and disposal.

Source

- MMSU, Hazardous Waste Disposal (SDG 15 evidence page) https://www.mmsu.edu.ph/resources/1jo6-3cVqxRAIlllxEhnSCgyT5ef5LGOW/data
- MMSU, Revised Integrated Pollution and Waste Management Program https://www.mmsu.edu.ph/resources/1fNUCZwyOozOKO27L2Bm6pNN-P_cJNpLd/view